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Analysis and research on quality control method of global radiation observation data

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Abstract: Radiation observation data are important basic data for assessment and research of climate change and solar energy resources. Quality control of radiation observation data is an important guarantee for accurate and reliable observation. First, the global radiation data collection method and the quality control process are introduced, and the differences and similarities between different quality control standards are compared in detail. Then, the quality control contents of the upper and lower limits and the rate of change are analysed theoretically and verified by experiments. Theoretical analysis and experimental results showed that the lower limit of the global radiation irradiance quality control should be −20 W/m^2, the upper limit should be 2221 W/m^2 or based on solar position, and the wrong change rate of sampling value should be 1000 W/m^2. Through the unification of the global radiation quality control standards, the difference caused by the inconsistent quality control methods can be avoided, thus improving the effectiveness of the radiation observation data and ensuring the accuracy and consistency of the radiation observation data.

1 Introduction

The global radiation in meteorology is defined as solar radiation component received from a solid angle of 2π steradian on a horizontal surface, which is an important variable in surface meteorological measurement, so it is widely measured in National Solar Radiation Networks, Global Atmospheric Watch Stations, and Baseline Surface Radiation Networks. In order to ensure the accuracy and reliability of the observation data, quality control of the global radiation observation data is usually required. In fact, at present, different kinds of quality control methods for global radiation measurements are presented in different standards and technical articles, such as the meteorological industry standard of ‘QX/T 66–2007 Specifications for Surface Meteorological Observation. Part22: Quality Control of Data’ [1], ‘QX/T 117–2010 Quality Control of Surface Radiation Observational Data’ [2], the national standard ‘GB/T 31156–2014 Solar Energy Resources Measurement – Global Radiation’ [3], ‘Function Specification Requirements of Automatic Weather Stations’ [4], ‘Function Specification Requirements of thermopile-based digital pyranometer’ [5] and ‘The Operational Manual of BASELINE SURFACE RADIATION NETWORK (BSRN)’ [6] and so on, various of quality control methods like the upper limit, lower limit, and change rate of global radiation are specified in these papers. However, some of quality control methods depicted in these norms and standards are not the same, and there are even contradictions in each other. Take the lower limit of global irradiance, for example, there are three different values of 0, −4, and −20 W/m^2 in different norms and standards, which is difficult for readers to understand and take use. Studies on the quality control method for global radiation have been carried out by many experts and scholars [7–9], but the researches are usually based on one of the certain methods mentioned above, applying the specified quality control method to global radiation observation data, making statistical analysis on such data, so there are few literatures related to analysis and research on the original method for global radiation quality control. Therefore, aiming at the inconsistent problem of global radiation quality control methods at present, the corresponding analysis and research to unify the quality control rules and methods for global radiation are of very important significance, which is the vital factor to ensure accuracy and reliability of the global radiation observation data used in the radiation climatology and scientific research.

In this article, a series of quality control processes for global radiation are introduced, the differences and similarities between different quality control methods are compared statistically. Based on the experiments and measurements, the key parameters of the lower limit, the upper limit, and the change rate of the global radiation are analysed, combining the different approaches, the lower limit of the global radiation quality control irradiance is −20 W/m^2, the upper limit is 2221 W/m^2 based on the solar position, and for real-time changes, the change rate of the wrong sampling value is 1000 W/m^2, such three constraint conditions will ensure the accuracy and reliability to some extent.

2 Observation and quality control methods for global radiation

2.1 Observation methods for global radiation

There are usually two variables in global radiation observation, which are irradiance and exposure. Irradiance represents intensity, mainly instantaneous intensity, while exposure represents the cumulative energy, including hourly exposure, daily exposure, monthly exposure etc. [10]

Irradiance is usually calculated from (1):

$$E = \frac{V}{K}, \tag{1}$$

where E is the global irradiance, unit is W·m^{-2}; V is the voltage output from the pyranometer, unit is μV; K is the sensitivity of the pyranometer, unit is μV·W^{-1}·m^2.

The exposure is usually calculated from (2):

$$I = \int_{t1}^{t2} (E \times t)\,\mathrm{d}t, \tag{2}$$

where I is the radiation exposure, unit is kJ·m^{-2} or (kW·h)·m^{-2}; E is the global irradiance, unit is W·m^{-2}; $t1$ and $t2$ are the start time or end time for pyranometer measurement, when $t2 - t1 = 1$ h, I is hourly exposure, when $t2 - t1 = 24$ h, I is daily exposure.

At present, in meteorological field in China, there are 98 surface radiation observation stations, four atmospheric watch stations, seven baseline surface radiation stations, and some local

Table 1 Global radiation observation in China

Station	Number	Instrument model	Remarks
surface radiation observation station	98	TBQ-2-B	first class, second class, third class
atmospheric watch station	4	PSP, CMP21	Shangdianzi, Lin'an, Longfengshan, Waliguan
baseline surface radiation station	7	CMP22, CMP21	Xilinhaote, Shangdianzi, Mohe, Xuchang, Wenjiang, Dali, Yanqi

Table 2 Summary of data collection methods of radiation observation

Sampling rate	Sampling interval, s	Calculation for instantaneous value	Related norms
6 per minute	10	remove the maximum and the minimum value, calculate the 1-minute average with the rest four values, and set the average as the instantaneous value for the minute	QX/T 61-2007 specifications for surface meteorological observation. Part 17: measurement at automatic meteorological station
30 per minute	2	calculate the average using the ‘correct’ samplings in one minute, and set the average as the instantaneous value for the minute	Function Specification Requirements of Automatic Weather Stations
30 per minute	2	calculate the average using the ‘correct’ samplings in one minute, and set the average as the instantaneous value for the minute	Function Specification Requirements of thermopile-based digital pyranometer
60 per minute	1	calculate the average using the ‘correct’ samplings in one minute, and set the average as the instantaneous value for the minute	The Operational Manual of Baseline Surface Radiation Network
6 per minute	10	remove the maximum and the minimum value, calculate the 1-minute average with the rest four values, and set the average as the instantaneous value for the minute	GB/T 31156-2014 Solar Energy Resources Measurement – Global Radiation

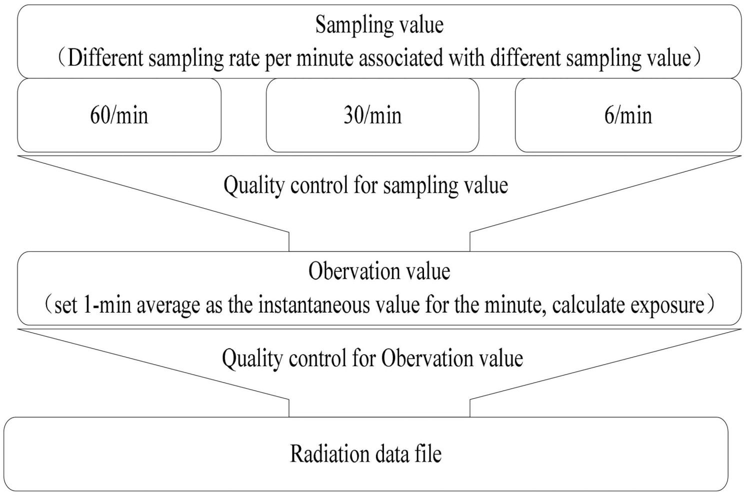

Fig. 1 *Two level quality control of radiation observation data*

Table 3 Quality control method of global radiation sampling value

Lower limit, W/m^2	Upper limit, W/m^2	Wrong change rate, W/m^2	Remarks
—	—	800	the operational manual of surface observation
0	2000	800	function specification requirements of automatic climatological stations
−20	2000	800	function specification requirements of thermopile-based digital pyranometer

self-built radiation observatories carrying out the global radiation measurement, the instruments, and the business distribution are listed in Table 1.

Refers to different kinds of stations, the collection method such as sampling rates is different; therefore, the algorithm used to calculate the global irradiance is not the same, the description for irradiance collection on different stations is shown in Table 2.

2.2 Radiation quality control procedure and methods

Essentially, the quality control procedure of radiation consists of two levels, the first level of the quality control is carried out for the sampling value of the sensor, and the second level is carried out for the observed value (usually the minute average of the sampling value) and the accumulated value, the procedure is shown in Fig. 1.

The quality control for sampling value includes the upper limit, lower limit, and the maximum permissible change rate, while the quality control of the observed instantaneous value and its accumulated value includes the upper limit, lower limit, doubtful change rate, and wrong change rate and so on.

The quality control for sampling value is conducted in the radiometric sensor, which is to control the quality of sampling values before calculating the observed instantaneous value using the sampling value, ensure that the sampling value used to calculate is correct. The main method involved in quality control of sampling values is shown in Table 3.

The quality control of radiation observation value (instantaneous value and accumulated value) usually contains climatological boundary value check, main variation range check,

Table 4 Climatological boundary check for radiation observation data

Parameters	Low limit	Upper limit	Doubtful change rate	Wrong change rate	Remarks
global irradiance		2000 W/m^2	800 W/m^2	1000 W/m^2	the operational manual of surface observation
	0 W/m^2	2000 W/m^2			QX/T 117-2010
	0 W/m^2	2000 W/m^2	800 W/m^2	1000 W/m^2	function specification requirements of automatic climatological stations
	−4 W/m^2	Sa × 1.5 × $\mu0^{1.2}$ + 100 W/m^2			the operational manual of baseline surface radiation network
		1600 W/m^2			GB/T 31156-2014
global exposure		(1 + 20%) R_{Dg}			the operational manual of surface observation
	0	(1 + 20%) R_{Dg}			QXT 117-2010
		(1 + 20%) R_{Dg}			GB/T 31156-2014

Note: Sa is the Solar constant divided by distance between the sun and the earth represented by astronomical units; RDg is the maximum daily radiation exposure in different altitude under sunny conditions.

Table 5 Irradiance upper limit

Physical extremum		Extremely rare	
Minimum	Maximum	Minimum	Maximum
0	2220.79	0	1746.63

and internal consistency check. The main methods are shown in Table 4.

3 Quality control method of global radiation irradiance

3.1 Lower limit quality control

At present, the 'lower limit' of the irradiance sampling value and the observed value has three values, that is 0, −4, and −20 W/m^2. We usually think that the irradiance value is caused by the solar radiation, and there is no negative value in theory. The causes of negative value of irradiance measurement is mainly related to the characteristics of the pyranometers used in current observation services.

The inductor of the pyranometer is composed of a circular (also square) blackened mica sheet and a thermocouple close to it. When the surface receives the global solar radiation, the thermocouple produces a temperature difference at both ends and produce a voltage output signal that is proportional to the solar radiation, and the irradiance can be calculated with the measurement of this voltage.

After sunset at night, there is no solar radiation, but due to the existence of 'cold sky', pyranometer will divert heat to the sky through the induction thermocouple, which is opposite to the solar radiation, and lead to the negative value of the radiant output. This phenomenon is called the 'Type A zero offset' [10] of the pyranometer. Type A zero offset is mostly between −20 and 0 W/m^2 [11]. Therefore, some technical standard specification define the lower limit of global radiation irradiance as −20 W/m^2.

In the actual radiation observation, because there is no solar radiation at night, and in order to avoid the negative value of irradiance, some acquisition software stops collecting data before sunset to sunrise, and sets the irradiance value to 0 W/m^2 directly. Therefore, the lowest measured value of irradiance is 0 W/m^2.

The zero offset of pyranometer is real existed. Although some radiometer has been revised to avoid the zero offset, but it cannot be completely eliminated. Moreover, the 'cold sky' is true, not only in the daytime or at night, but because of the radiation intensity of the sun in the daytime, the 'cold source' is inundated. It is not appropriate to stop collecting data at night and set the irradiance to 0 W/m^2 directly either.

Therefore, it is suggested to set the 'lower limit' of the global irradiance quality to −20 W/m^2.

3.2 'Upper limit' quality control

At present, the 'upper limit' of irradiance quality control is 2000 and 1600 W/m^2. The 'upper limit' of quality control in the BSRN gives two values, which is physical extremes value and extremely rare value, and all are variables, which need to be calculated according to the zenith angle Z and the distance between the sun and the earth AU, as shown in formula (3) and formula (4).

$$E_{\max} = \frac{S_0}{\mathrm{AU}^2} \times 1.5 \times \cos(z)^{1.2} + 100\,\mathrm{W/m^2} \tag{3}$$

$$E_{\mathrm{extreme}} = \frac{S_0}{\mathrm{AU}^2} \times 1.2 \times \cos(z)^{1.2} + 50\,\mathrm{W/m^2} \tag{4}$$

where $E_{\max}$ – Irradiance upper limit physical extremum, unit W·m^{-2}; E_{extreme} – The upper limit of irradiance is extremely rare, unit W·m^{-2}; S_0 – Solar constant, $S_0 = 1367\mathrm{W/m^2}$; AU – Distance between the sun and the earth represented by astronomical units; z – Sun zenith angle.

As Z and AU changes at all times, the upper limit is also changing in every time. The variation of zenith angle Z is 0∼90 degrees. When the zenith angle is $Z = 0°$ and solar altitude angle is 90°, the sun is vertical. The average distance between the sun and the earth is 1.496 × 108 km, an astronomical unit 1 AU. The minimum distance between the sun and the earth is about 1.471 × 108 km, about 0.98 AU, and the maximum distance between the sun and the earth is about 1.521 × 108 km, about 1.02 AU.

Considering the extreme position of the sun, the maximum value of the 'upper limit' is when the zenith angle $Z = 0°$ and the sun earth distance 0.98 AU. When the zenith angle $Z = 90°$ and the sun earth distance 1.02 AU are the minimum values of the 'upper limit'. As a result, the physical extremum of 'upper limit' is changed between 0∼2221 W/m^2, and the extremely rare of 'upper limit' is changed between 0∼1747 W/m^2, as shown in Table 5.

Therefore, considering the limit of quality control, it is suggested to set the upper limit of global radiation irradiance to 2221 W/m^2.

If the observation business capability is allowed, the 'upper limit' of the radiation quality control varies according to the change of the solar altitude angle and the sun earth distance in accordance with the standard of observation of BSRN.

Table 6 Change rate pyranometer

No.	Type	ID	Sensitivity	Response time, s
1	CMP22	060016	8.96	5
2	TBQ-2-B	2326	12.22	19

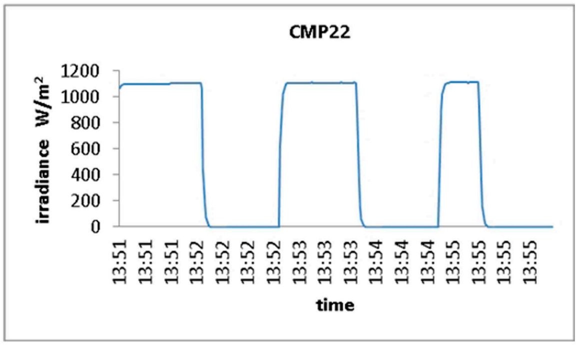

Fig. 2 *Output curve of CMP22*

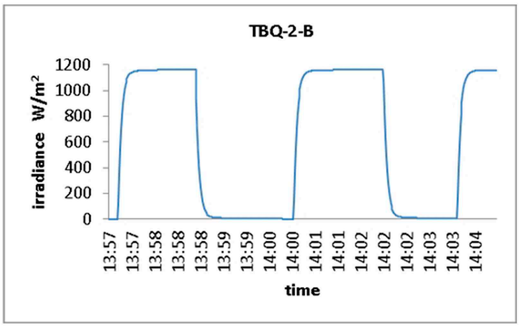

Fig. 3 *Output curve of TBQ-2-B*

Table 7 change rate test result

Pyranometer	Variation extremum within 1s, 60/min	Variation extremum within 2s, 30/min	Variation extremum within 10s, 6/min
CMP22	611 W/m^2	899 W/m^2	1111 W/m^2
TBQ-2-B	235 W/m^2	447 W/m^2	1052 W/m^2

3.3 'Change rate' quality control

The 'change rate' quality control refers to the maximum allowable change between two adjacent values. At present, in the 'change rate' quality control, there is a slight difference between the sampling value and the observed value for the 'change rate'. The sampling value specifies that the error rate of change is 800 W/m^2. The observed rate set the rate of doubt change is 800 W/m^2, and the erroneous change rate is 1000 W/m^2.

In solar radiation observation, there are three main sampling frequencies, six times per minute, 30 times per minute and 60 times per minute. The fastest sampling frequency is one second and one sampling [12, 13]. In order to understand the limit of change rate, we have prepared two pyranometers for test, the result is shown in Table 6. At the 1200 W/m^2 of solar simulator in National Meteorological Metrology Centre in China, the change rate of change irradiance is analysed by the method of 'cover and uncover'.

The output curve of CMP22 is shown in Fig. 2, and the output curve of TBQ-2-B is shown in Fig. 3.

The change rate of the two pyranometer sampling values is shown in Table 7.

From the experiment, we can see that the rate of change is different at different sampling frequencies. If the sampling frequency is 60 times/min, the maximum rate of change between the two sampling values do not exceed 800 W/m^2. If the sampling frequency is 30 times per min, the maximum change rate between the two sampling values of CMP22 is 899 W/m^2, which exceeds 800 W/m^2. If the sampling frequency is six times per min, whether CMP22 or TBQ-2-B, the maximum rate of change between their two sampling values exceeds 800 W/m^2.

Of course, the experiment was done under the irradiation of the simulated solar source of the solar simulator. The 'shaded and unshaded' experiment was carried out instantaneously. In reality, there will be 'shaded and unshaded ' phenomenon of clouds. But due to the existence of scattered radiation, the change of irradiance may not be as dramatic as that in the laboratory. However, the quality control of 'change rate' is generally considered in the extreme case [14, 15]. Taking all factors into consideration, it is suggested that the 'change rate' quality control standard of sampling error should be set to 1000 W/m^2. Or, according to different sampling frequency, adopt different 'change rate' quality control standard.

4 Quality control method of global radiation exposure

4.1 'Lower limit' of t global radiation exposure

In the current quality control requirements, only the *QX/T 117–2010 surface meteorological radiation observation data quality control* gives the global radiation daily exposure 'lower limit', which is 0. No special description is made in other standards. Daily radiation exposure value comes from the integral calculation of irradiance, which should not be <0. Therefore, the 'lower limit' of global radiation daily exposure can be determined to be 0.

4.2 'Upper limit' of t global radiation exposure

In the current quality control requirements, the 'upper limit' of the global radiation daily exposure is based on the maximum global radiation daily RDg of possible global radiant radiation in all latitudes under sunny conditions. Considering the convenience, feasibility, and continuity of quality control, it is recommended to set the 'upper limit' of global radiation daily exposure to (1 + 20%) RDg.

5 Conclusion

The quality control of global radiation includes two aspects: irradiance and exposure. The contents of quality control mainly include the upper and lower limits and the rate of change. The quality control process is, first of all, the quality control of the sampled values, and then the quality control of the observed values and their accumulated values. On the basis of analysing the similarities and differences of different specifications and related experiments, it is suggested that the unified global radiation quality control standards be as follows: The lower limit of global radiation irradiance is −20 W/m^2, the upper limit is 2221 W/m^2 or calculated by certain formula. The error change rate of the sampling value is 1000 W/m^2. As of the big amount of observation involved in radiation, which includes direct radiation, scattering radiation, reflected radiation, long-wave radiation, and so on. The quality control of these radiation needs further study and unification.

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